Step 1: Update Ubuntu and Install Java (JDK)

Hadoop is Java-based, so this is the most critical dependency.

1. Open your Ubuntu terminal.
2. Update the package list:

Bash

sudo apt update && sudo apt upgrade -y

3. Install OpenJDK 8 (Hadoop works most reliably with Java 8 or 11):

Bash

sudo apt install openjdk-8-jdk -y

4. Verify the installation:

Bash

java -version

Step 2: Configure SSH (Passwordless Login)

Hadoop uses SSH to manage its nodes (even on a single machine). You need to allow it to connect to "localhost" without asking for a password every time.

1. Install OpenSSH Server:

Bash

sudo apt install openssh-server openssh-client -y

2. Generate a key pair (press Enter for all prompts):

Bash

ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa

3. Add the key to authorized keys:

Bash

cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

chmod 0600 ~/.ssh/authorized_keys

4. Test it (if it asks for a password, something went wrong):

Bash

ssh localhost

(Type exit to return after testing).

Step 3: Download and Extract Hadoop

1. Go to the Apache Hadoop Releases page or use wget to download version 3.3.6 (the current stable version):

Bash

wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz

2. Extract the file:

Bash

tar -xvzf hadoop-3.3.6.tar.gz

3. Move it to a standard directory:

Bash

sudo mv hadoop-3.3.6 /usr/local/hadoop

Step 4: Set Environment Variables

You need to tell Ubuntu where Java and Hadoop live.

1. Open your bash profile:

Bash

nano ~/.bashrc

2. Scroll to the bottom and paste the following:

Bash

Java Home

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

Hadoop Home

export HADOOP_HOME=/usr/local/hadoop

export HADOOP_INSTALL=$HADOOP_HOME

export HADOOP_MAPRED_HOME=$HADOOP_HOME

export HADOOP_COMMON_HOME=$HADOOP_HOME

export HADOOP_HDFS_HOME=$HADOOP_HOME

export YARN_HOME=$HADOOP_HOME

export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native

export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin

export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

3. Save and Exit (**Ctrl+O**, **Enter**, **Ctrl+X**).
4. Apply the changes:

Bash

source ~/.bashrc

Step 5: Configure Hadoop Files

Hadoop requires specific settings in its XML files to run in "Pseudo-Distributed" mode.

1. hadoop-env.sh

Tell Hadoop where Java is.

Bash

nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

Find the line export JAVA_HOME=... and set it to: export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

2. core-site.xml

Define the NameNode location.

Bash

```
nano $HADOOP_HOME/etc/hadoop/core-site.xml
```

Paste this inside the <configuration> tags:

XML

```
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://localhost:9000</value>
</property>
```

3. hdfs-site.xml

Define how many times data is copied (1 for local testing).

Bash

```
nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

Paste this inside the <configuration> tags:

XML

```
<property>
  <name>dfs.replication</name>
  <value>1</value>
</property>
```

Step 6: Format and Start Hadoop

Before running it for the first time, you must format the file system.

1. **Format NameNode:**

Bash

```
hdfs namenode -format
```

2. **Start HDFS (NameNode and DataNode):**

Bash

start-dfs.sh

3. **Start YARN (Resource Manager and Node Manager):**

Bash

start-yarn.sh

4. **Verify if they are running:** Type jps. You should see NameNode, DataNode, ResourceManager, and NodeManager in the list.

If not root

1. **Open your own bash config:**

Bash

nano ~/.bashrc

2. **Scroll to the bottom and paste the same block we used earlier:**

Bash

Java Home

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

Hadoop Home

export HADOOP_HOME=/usr/local/hadoop

export HADOOP_INSTALL=$HADOOP_HOME

export HADOOP_MAPRED_HOME=$HADOOP_HOME

export HADOOP_COMMON_HOME=$HADOOP_HOME

export HADOOP_HDFS_HOME=$HADOOP_HOME

export YARN_HOME=$HADOOP_HOME

export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native

export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin

export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"

3. **Save and Exit:** (**Ctrl+O**, **Enter**, **Ctrl+X**).
4. **Refresh the settings:**

Bash

source ~/.bashrc

---

2. Update Hadoop Env for the New User

Earlier, we told Hadoop to run as "root." Now that you are "shreya," we need to update that in the configuration file, or Hadoop will try to start services as the wrong user.

1. **Open the hadoop-env.sh file:**

Bash

nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

2. **Find the lines we added at the bottom and change "root" to "shreya":**

Bash

export HDFS_NAMENODE_USER="shreya"

export HDFS_DATANODE_USER="shreya"

export HDFS_SECONDARYNAMENODE_USER="shreya"

export YARN_RESOURCEMANAGER_USER="shreya"

export YARN_NODEMANAGER_USER="shreya"

3. **Save and Exit.**

---

3. Try Starting Again

Now, try the commands again:

Bash

start-dfs.sh

start-yarn.sh

Then check with:

Bash

jps

1. Open the file

Bash

nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

2. Add these lines at the end

Make sure each one is on its own line:

Bash

export HDFS_NAMENODE_USER="shreya"

export HDFS_DATANODE_USER="shreya"

export HDFS_SECONDARYNAMENODE_USER="shreya"

export YARN_RESOURCEMANAGER_USER="shreya"

export YARN_NODEMANAGER_USER="shreya"

WHEN YOU ADD USER

root@ShreyaYadav:~# adduser hadoop

root@ShreyaYadav:~# usermod -aG sudo hadoo

root@ShreyaYadav:~# su – hadoop

root@ShreyaYadav:~# chown -R shreya:shreya /usr/local/hadoop

root@ShreyaYadav:~# su - shreya

shreya@ShreyaYadav:~$ start-dfs.sh

start-dfs.sh: command not found

```
shreya@ShreyaYadav:~$ nano ~/.bashrc

shreya@ShreyaYadav:~$ sourcde ~/.bashrc

sourcde: command not found

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

shreya@ShreyaYadav:~$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa

Generating public/private rsa key pair.

Created directory '/home/shreya/.ssh'.

Your identification has been saved in /home/shreya/.ssh/id_rsa

Your public key has been saved in /home/shreya/.ssh/id_rsa.pub

The key fingerprint is:

SHA256:vuVwRXy/hcEUTa1KDu2PpVSXaD4opzXPlgq3glo4b+M shreya@ShreyaYadav

The key's randomart image is:

+---[RSA 3072]----+

|          o+o|

|        . o  o|

|        .o =..|

|       ..o+o=.|

|      S  ==o..o|

|     .. . *=o. o|

|    o .+.Bo+=o. |

|     ++ Oo +=.  |

|    .+Eo ooo    |

+----[SHA256]-----+

shreya@ShreyaYadav:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys

chmod 0600 ~/.ssh/authorized_keys
```

shreya@ShreyaYadav:~$ ssh localhost

The authenticity of host 'localhost (127.0.0.1)' can't be established.

ED25519 key fingerprint is SHA256:Dx//8WA11fGv16JJ9Y43mVrnhmNXpfgFQ2mgZf2Txvc.

This key is not known by any other names

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

* Documentation: https://help.ubuntu.com

* Management: https://landscape.canonical.com

* Support: https://ubuntu.com/pro

System information as of Sun Apr 26 12:00:39 IST 2026

System load: 0.08   Processes: 72

Usage of /:   2.1% of 1006.85GB   Users logged in: 1

Memory usage: 44%   IPv4 address for eth0: 192.168.38.30

Swap usage: 9%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s

just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

New release '24.04.4 LTS' available.

Run 'do-release-upgrade' to upgrade to it.

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

*** Documentation: https://help.ubuntu.com**

*** Management: https://landscape.canonical.com**

*** Support: https://ubuntu.com/pro**

System information as of Sun Apr 26 02:29:51 IST 2026

System load: 2.08 Processes: 49

Usage of /: 2.0% of 1006.85GB Users logged in: 1

Memory usage: 38% IPv4 address for eth0: 192.168.38.30

Swap usage: 1%

*** Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s**

just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

New release '24.04.4 LTS' available.

Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Apr 16 20:50:42 2026

shreya@ShreyaYadav:~$ start-dfs.sh

Starting namenodes on [localhost]

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting datanodes

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting secondary namenodes [ShreyaYadav]

ShreyaYadav: Warning: Permanently added 'shreyayadav' (ED25519) to the list of known hosts.

ShreyaYadav: ERROR: JAVA_HOME is not set and could not be found.

shreya@ShreyaYadav:~$ start-yarn.sh

Starting resourcemanager

Starting nodemanagers

localhost: ERROR: JAVA_HOME is not set and could not be found.

shreya@ShreyaYadav:~$ readlink -f $(which java) | sed "s:/bin/java::"

/usr/lib/jvm/java-8-openjdk-amd64/jre

shreya@ShreyaYadav:~$ nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ start-dfs.sh

Starting namenodes on [localhost]

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting datanodes

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting secondary namenodes [ShreyaYadav]

ShreyaYadav: ERROR: JAVA_HOME is not set and could not be found.

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ nano ~/.bashrc

shreya@ShreyaYadav:~$ nano ~/.bashrc

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ start-dfs.sh

Starting namenodes on [localhost]

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting datanodes

localhost: ERROR: JAVA_HOME is not set and could not be found.

Starting secondary namenodes [ShreyaYadav]

ShreyaYadav: ERROR: JAVA_HOME is not set and could not be found.

shreya@ShreyaYadav:~$ nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh

shreya@ShreyaYadav:~$ source ~/.bashrc

shreya@ShreyaYadav:~$ start-dfs.sh

Starting namenodes on [localhost]

Starting datanodes

Starting secondary namenodes [ShreyaYadav]

shreya@ShreyaYadav:~$ start-yarn.sh

Starting resourcemanager

resourcemanager is running as process 63415. Stop it first and ensure /tmp/hadoop-shreya-resourcemanager.pid file is empty before retry.

Starting nodemanagers

shreya@ShreyaYadav:~$ jps

65040 DataNode

63415 ResourceManager

65240 SecondaryNameNode

65611 NodeManager

65786 Jps

shreya@ShreyaYadav:~$ stop-all.sh

WARNING: Stopping all Apache Hadoop daemons as shreya in 10 seconds.

WARNING: Use CTRL-C to abort.

Stopping namenodes on [localhost]

Stopping datanodes

Stopping secondary namenodes [ShreyaYadav]

Stopping nodemanagers

Stopping resourcemanager

shreya@ShreyaYadav:~$ hdfs namenode -format

shreya@ShreyaYadav:~$ start-dfs.sh

Starting namenodes on [localhost]

Starting datanodes

Starting secondary namenodes [ShreyaYadav]

shreya@ShreyaYadav:~$ start-yarn.sh

Starting resourcemanager

Starting nodemanagers

shreya@ShreyaYadav:~$ jps

68003 Jps

66949 DataNode

67478 ResourceManager

66809 NameNode

67608 NodeManager

67194 SecondaryNameNode

shreya@ShreyaYadav:~$ stop-all.sh

WARNING: Stopping all Apache Hadoop daemons as shreya in 10 seconds.

WARNING: Use CTRL-C to abort.

Stopping namenodes on [localhost]

Stopping datanodes

Stopping secondary namenodes [ShreyaYadav]

Stopping nodemanagers

Stopping resourcemanager

shreya@ShreyaYadav:~$ mkdir ~/mapreduce

shreya@ShreyaYadav:~$ cd ~/mapreduce